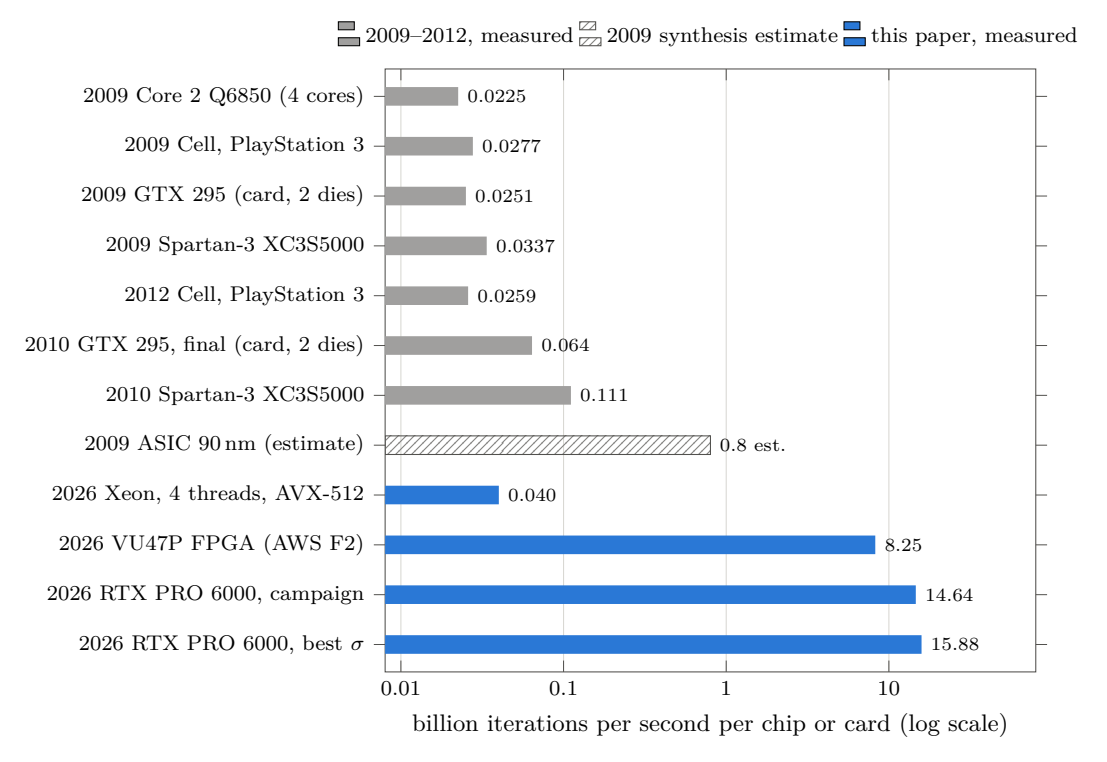
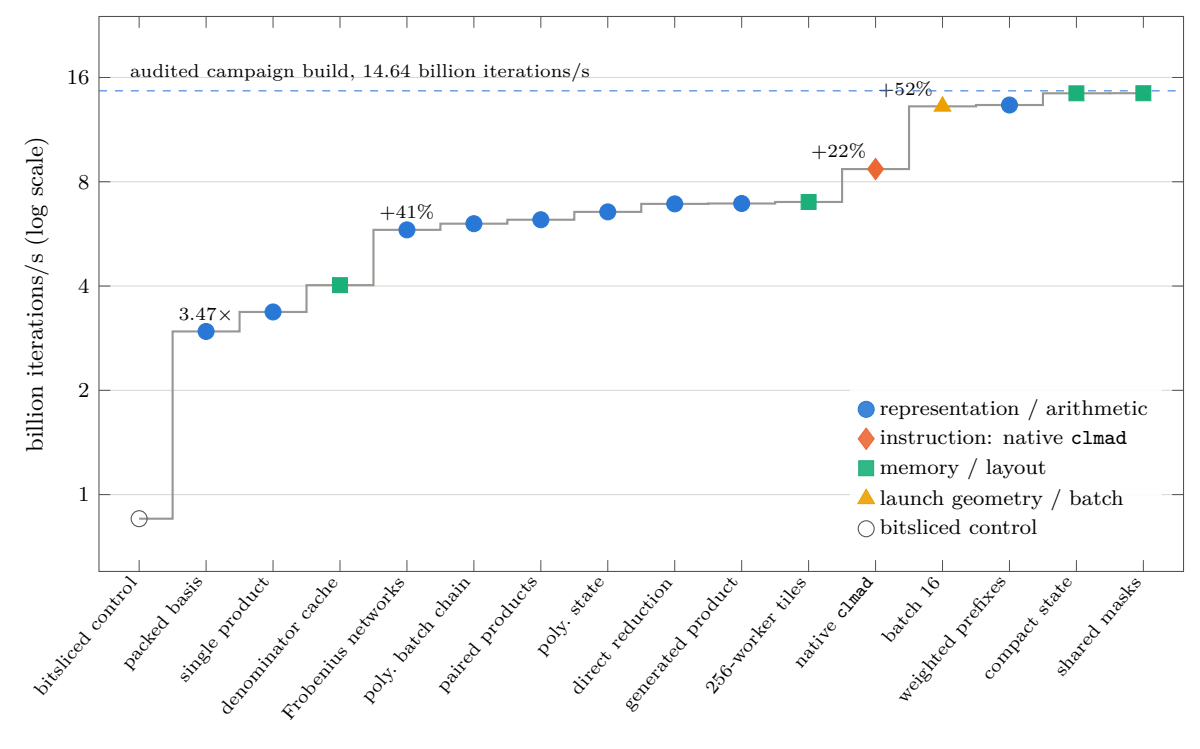
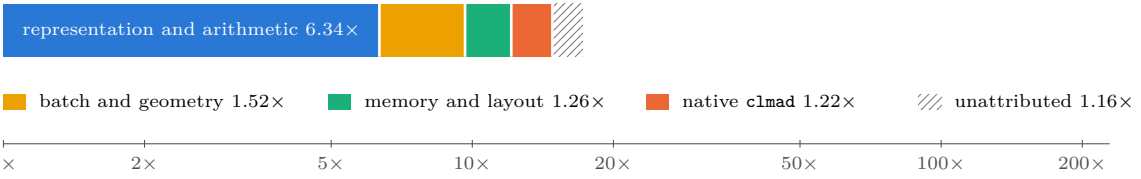

2009–2012, measured
2009 synthesis estimate
this paper, measured
2009 Core 2 Q6850 (4 cores)
0.0225
2009 Cell, PlayStation 3
0.0277
2009 GTX 295 (card, 2 dies)
0.0251
2009 Spartan-3 XC3S5000
0.0337
2012 Cell, PlayStation 3
0.0259
2010 GTX 295, final (card, 2 dies)
0.064
2010 Spartan-3 XC3S5000
0.111
2009 ASIC 90 nm (estimate)
0.8 est.
2026 Xeon, 4 threads, AVX-512
0.040
2026 VU47P FPGA (AWS F2)
8.25
2026 RTX PRO 6000, campaign
14.64
2026 RTX PRO 6000, best σ
15.88
0.01
0.1
1
10
billion iterations per second per chip or card (log scale)


audited campaign build, 14.64 billion iterations/s
billion iterations/s (log scale)
16
8
4
2
1
3.47×
+41%
+22%
+52%
representation / arithmetic
instruction: native clmad
memory / layout
launch geometry / batch
bitsliced control
bitsliced control
packed basis
single product
denominator cache
Frobenius networks
poly. batch chain
paired products
poly. state
direct reduction
generated product
256-worker tiles
native clmad
batch 16
weighted prefixes
compact state
shared masks

Against the 2010 GTX 295 card: 229×




The ledger, bitsliced control to audited build: 17.2×



## Grid: 1,504 blocks of 256 threads, run as four waves

| wave 1: 376 blocks | wave 2: 376 blocks | wave 3: 376 blocks | wave 4: 376 blocks |
|---|---|---|---|

385,024 workers = 4× the 96,256 that fit at once (188 SMs × 2 blocks × 256 threads)

### One SM (of 188)

block 0: 256 threads

block 1: 256 threads

shared: 1,792-byte $\sigma$ mask table per block

16 warps per SM; 104 registers per thread

### One thread = one worker = 16 walks

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |

walks 0–15: one field inversion per step serves all 16 (Montgomery's trick)

### Where the state lives

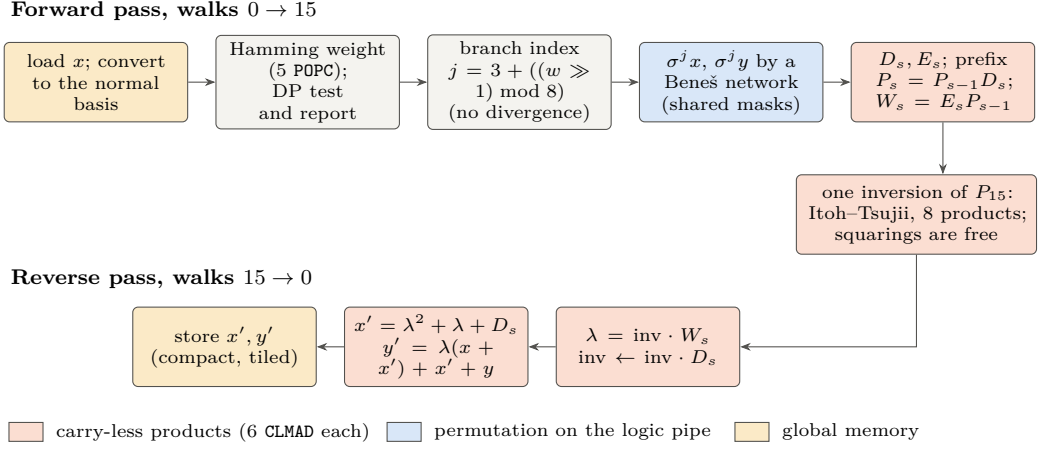
**registers**: running product and inverse; the current walk's $x, y, D, E, \lambda$; unreduced products

**shared / constant**: $\sigma$ masks (1,792 B); inverse-network masks; start-point tables

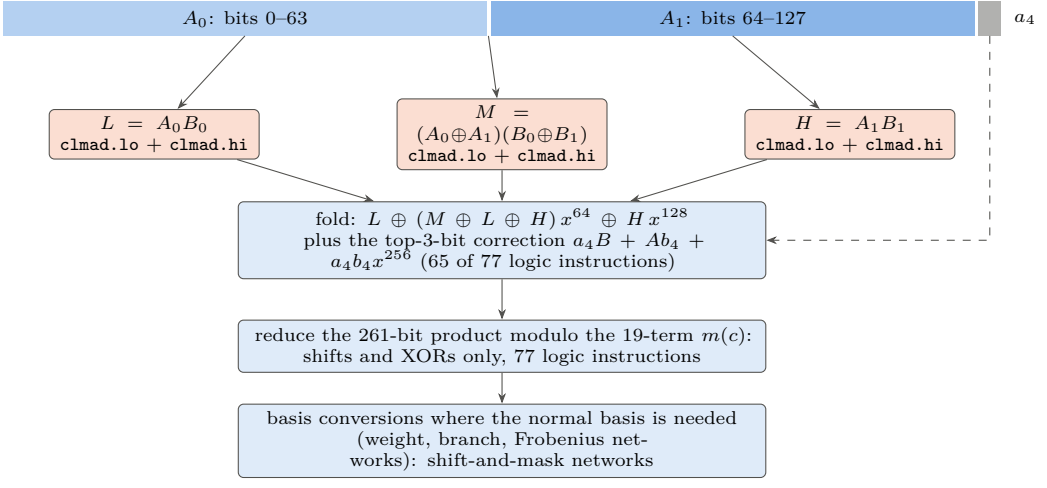
**global**: $x$, $y$, prefix products and cached denominators, 17 B per element in 256-worker tiles

**distinguished points**: `atomicAdd` into a device buffer, copied to the host between launches

**checkpoint**: 60 bytes per walk, about 370 MB for the whole grid


Forward pass, walks $0 \to 15$
load $x$; convert to the normal basis
Hamming weight (5 POPC); DP test and report
branch index $j = 3 + ((w \gg 1) \bmod 8)$ (no divergence)
$\sigma^j x$, $\sigma^j y$ by a Beneš network (shared masks)
$D_s, E_s$; prefix $P_s = P_{s-1} D_s$; $W_s = E_s P_{s-1}$
one inversion of $P_{15}$: Itoh–Tsujii, 8 products; squarings are free
Reverse pass, walks $15 \to 0$
$\lambda = \text{inv} \cdot W_s$ $\text{inv} \leftarrow \text{inv} \cdot D_s$
$x' = \lambda^2 + \lambda + D_s$ $y' = \lambda(x + x') + x' + y$
store $x', y'$ (compact, tiled)
carry-less products (6 CLMAD each)
permutation on the logic pipe
global memory

**A 131-bit operand in five 32-bit words, read as two 64-bit limbs and 3 top bits**



**Untiled: one array per (slot, word) spanning all 385,024 workers**




a block's 256 threads touch 80 regions, each 1.5 MB apart

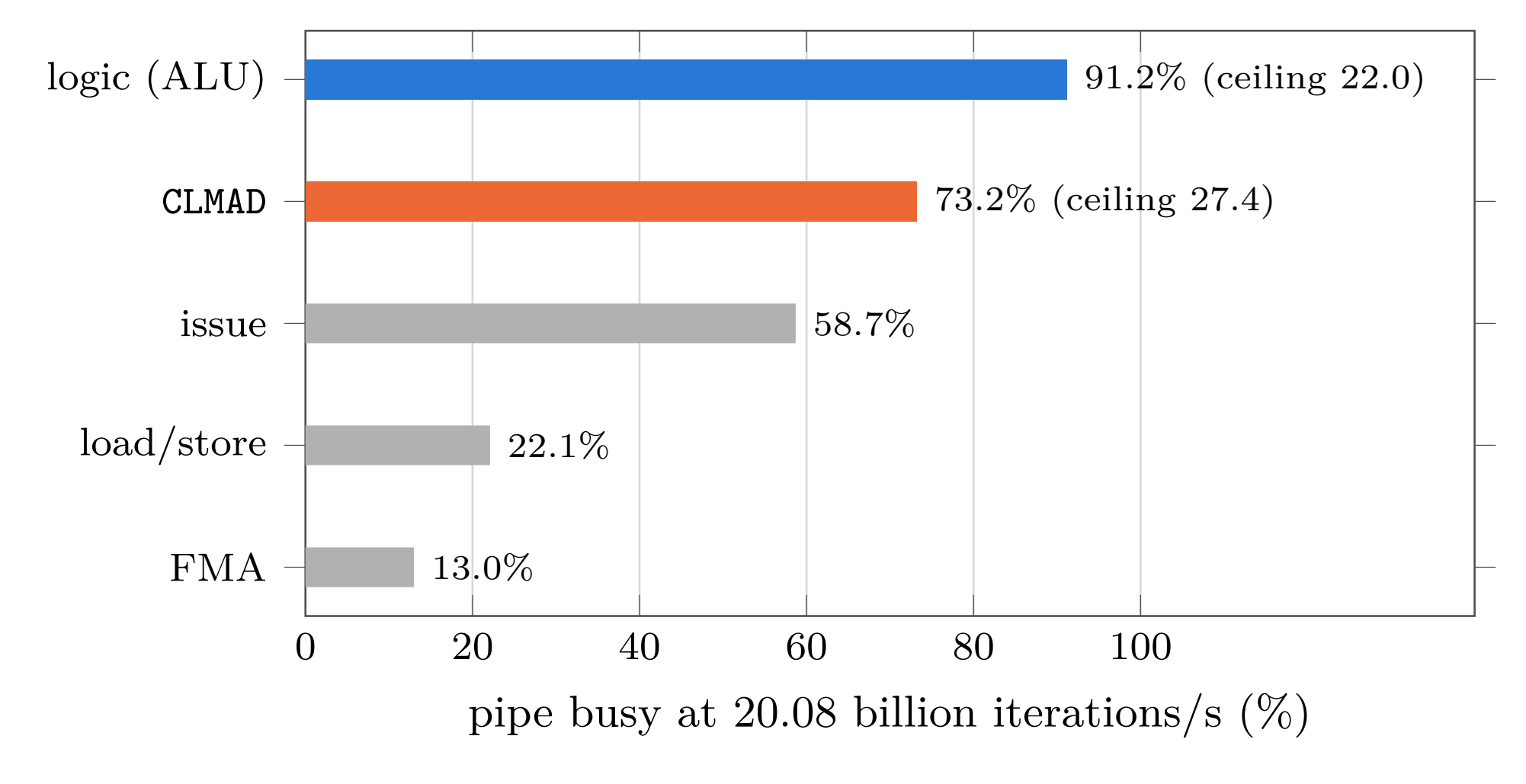
**Tiled: each block's 256 workers keep all their slots and words together**



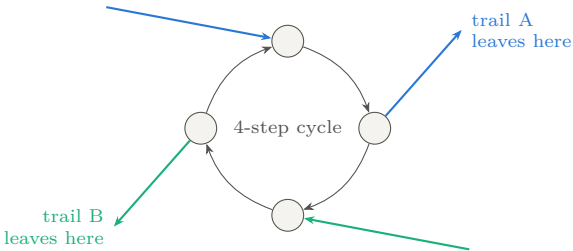

**Compact element record for one (tile, slot): 4,352 bytes**




17 bytes per element instead of 20: the top 3 bits (or 6, with the cached branch) share one byte; the four state arrays fall from 470 to 399.5 MiB.

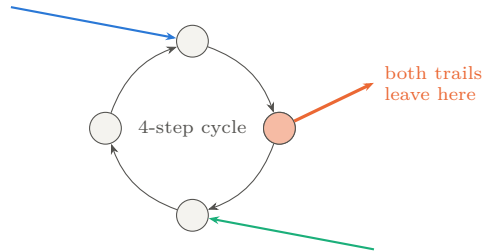

logic (ALU)
CLMAD
issue
load/store
FMA
91.2% (ceiling 22.0)
73.2% (ceiling 27.4)
58.7%
22.1%
13.0%
0
20
40
60
80
100
pipe busy at 20.08 billion iterations/s (%)

**Rule v2: the exit depends on the history**

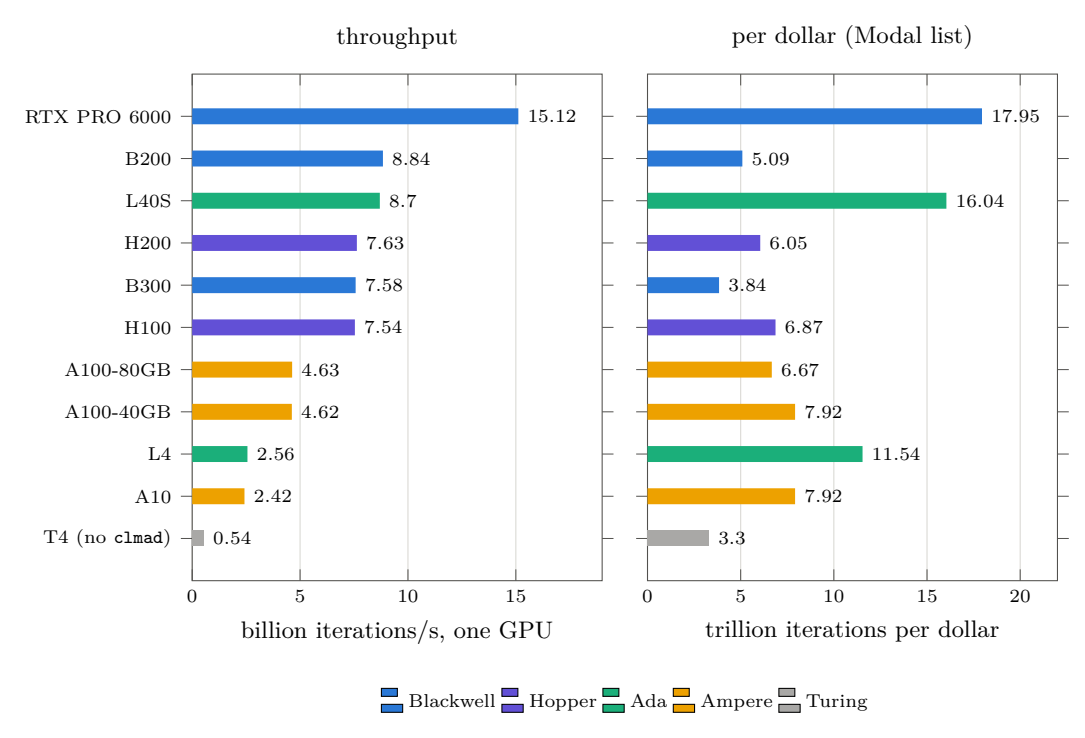


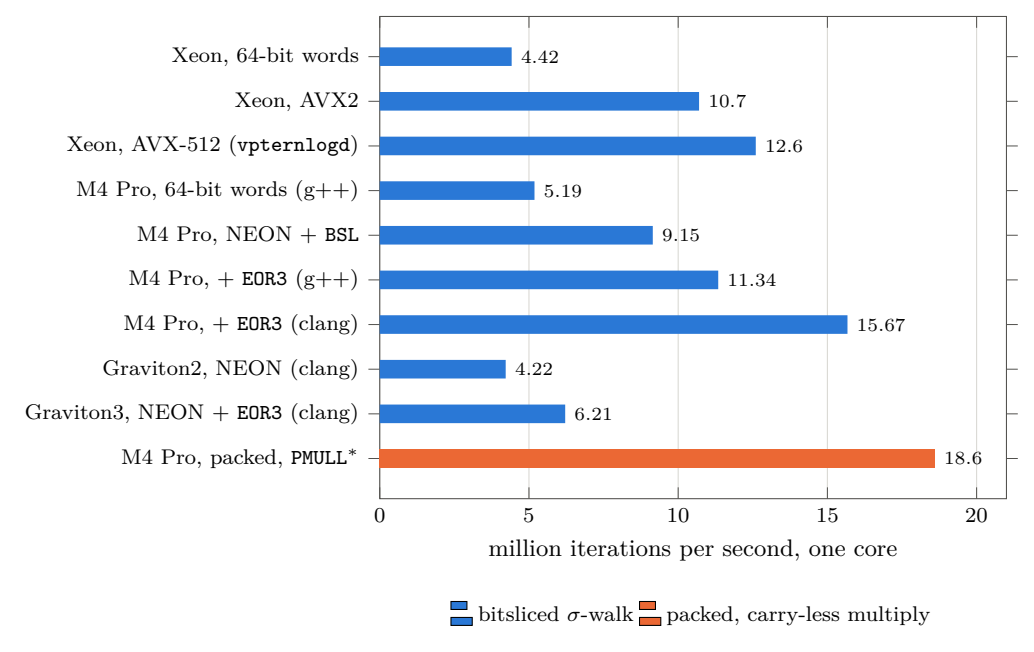

the two trails need not merge,
so their collision can be missed

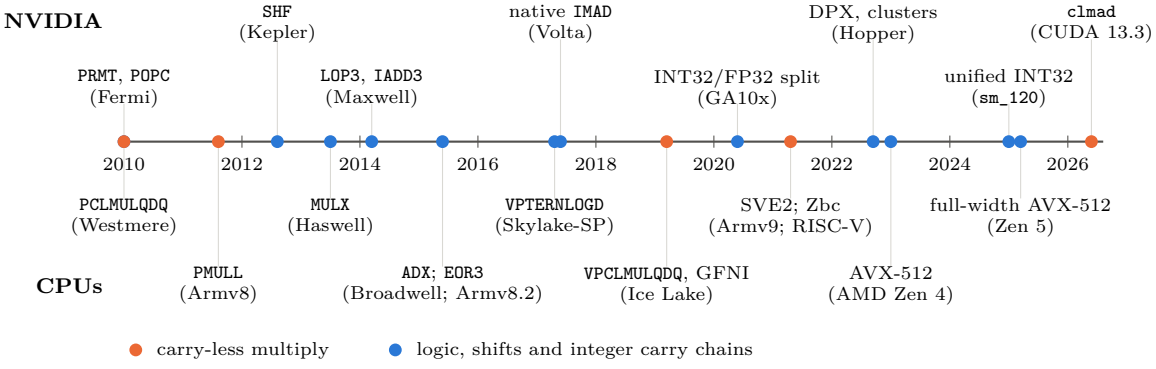
**Rule v3: one anchor per cycle**

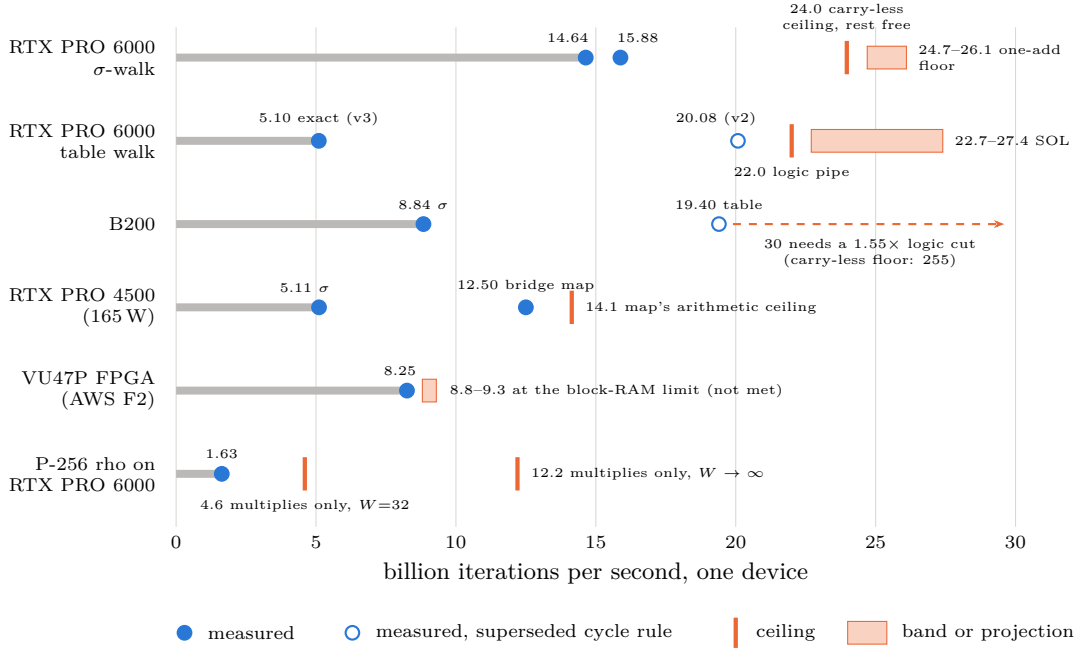



exit chosen from the cycle itself
(least weight and $x$), so trails coalesce


throughput
per dollar (Modal list)
RTX PRO 6000
B200
L40S
H200
B300
H100
A100-80GB
A100-40GB
L4
A10
T4 (no clmad)
15.12
8.84
8.7
7.63
7.58
7.54
4.63
4.62
2.56
2.42
0.54
17.95
5.09
16.04
6.05
3.84
6.87
6.67
7.92
11.54
7.92
3.3
0
5
10
15
20
billion iterations/s, one GPU
trillion iterations per dollar
Blackwell
Hopper
Ada
Ampere
Turing


Xeon, 64-bit words
4.42
Xeon, AVX2
10.7
Xeon, AVX-512 (vpternlogd)
12.6
M4 Pro, 64-bit words (g++)
5.19
M4 Pro, NEON + BSL
9.15
M4 Pro, + EOR3 (g++)
11.34
M4 Pro, + EOR3 (clang)
15.67
Graviton2, NEON (clang)
4.22
Graviton3, NEON + EOR3 (clang)
6.21
M4 Pro, packed, PMULL*
18.6
0
5
10
15
20
million iterations per second, one core
bitsliced σ-walk
packed, carry-less multiply


NVIDIA
SHF
(Kepler)
native IMAD
(Volta)
DPX, clusters
(Hopper)
clmad
(CUDA 13.3)
PRMT, POPC
(Fermi)
LOP3, IADD3
(Maxwell)
INT32/FP32 split
(GA10x)
unified INT32
(sm_120)
2010
2012
2014
2016
2018
2020
2022
2024
2026
PCLMULQDQ
(Westmere)
MULX
(Haswell)
VPTERNLOGD
(Skylake-SP)
SVE2; Zbc
(Armv9; RISC-V)
full-width AVX-512
(Zen 5)
CPUs
PMULL
(Armv8)
ADX; EOR3
(Broadwell; Armv8.2)
VPCLMULQDQ, GFNI
(Ice Lake)
AVX-512
(AMD Zen 4)
carry-less multiply
logic, shifts and integer carry chains


RTX PRO 6000
σ-walk
14.64
15.88
24.0 carry-less
ceiling, rest free
24.7–26.1 one-add
floor
RTX PRO 6000
table walk
5.10 exact (v3)
20.08 (v2)
22.7–27.4 SOL
22.0 logic pipe
B200
8.84 σ
19.40 table
30 needs a 1.55× logic cut
(carry-less floor: 255)
RTX PRO 4500
(165 W)
5.11 σ
12.50 bridge map
14.1 map's arithmetic ceiling
VU47P FPGA
(AWS F2)
8.25
8.8–9.3 at the block-RAM limit (not met)
P-256 rho on
RTX PRO 6000
1.63
4.6 multiplies only, W=32
12.2 multiplies only, W → ∞
0
5
10
15
20
25
30
billion iterations per second, one device
measured
measured, superseded cycle rule
ceiling
band or projection